

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

CHEMICAL COORDINATION & INTEGRATION

LECTURE-02

ZOOLOGY

DAVOOD SIR (CEO of NG)

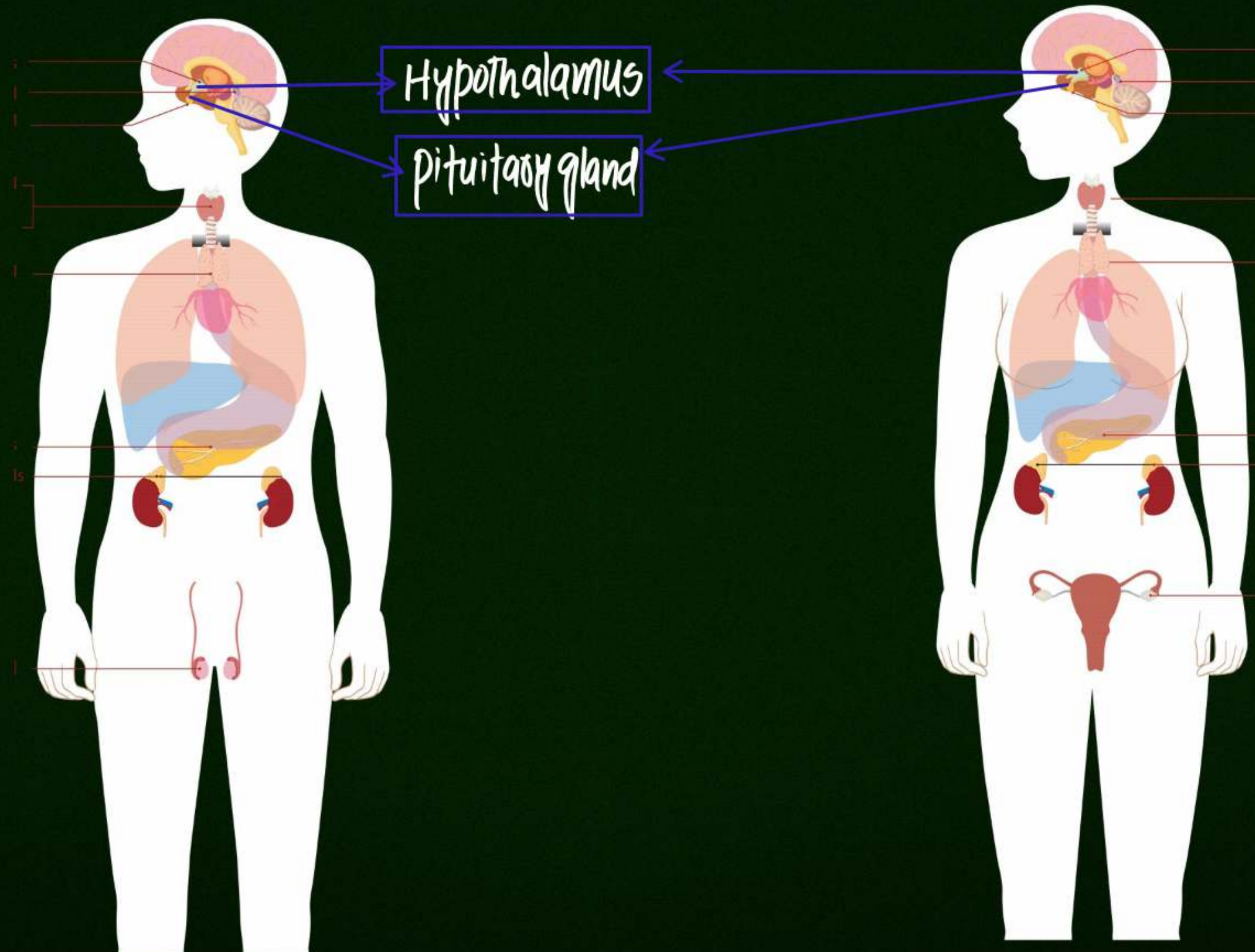


Physics Wallah

Topics To Be Covered

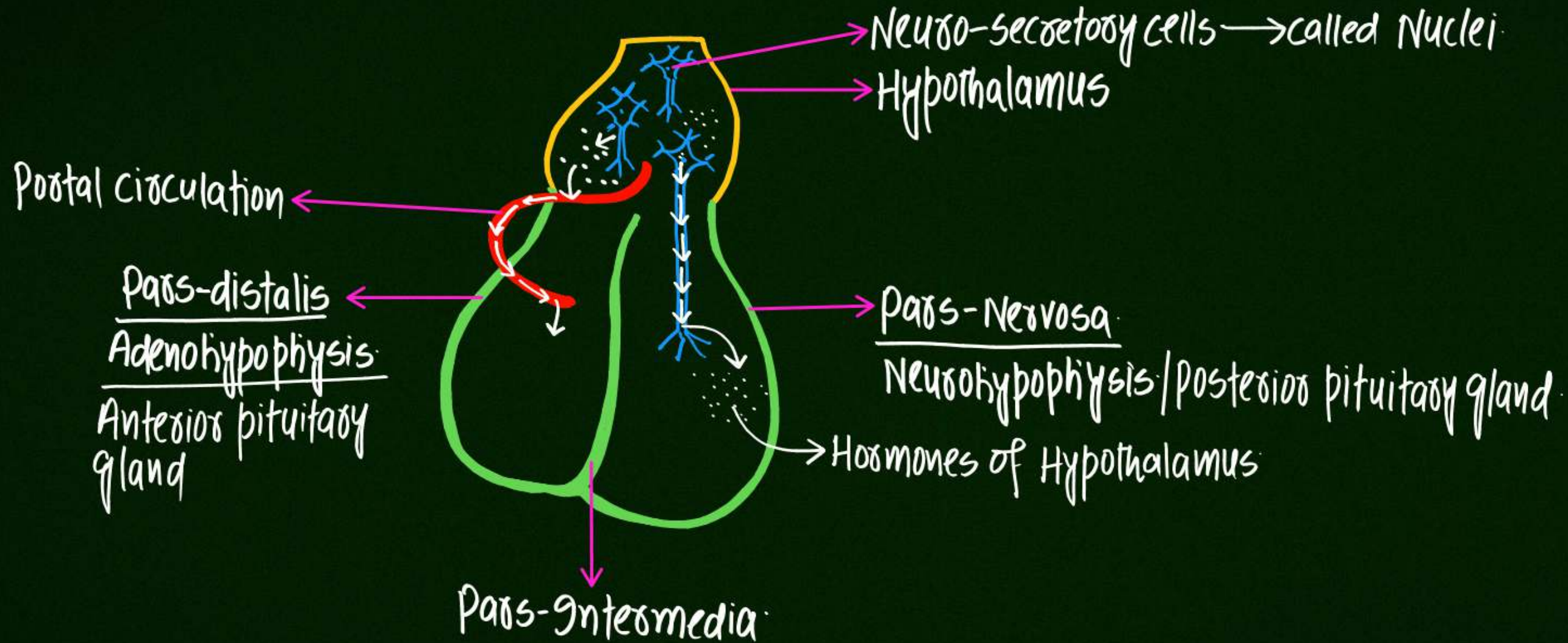
- 1 A white rectangular box with a red border, containing a redacted white shape.
- 2 HUMAN ENDOCRINE SYSTEM.
- 3 PREVIOUS & CURRENT YEAR NEET MCQ's

Human Endocrine System

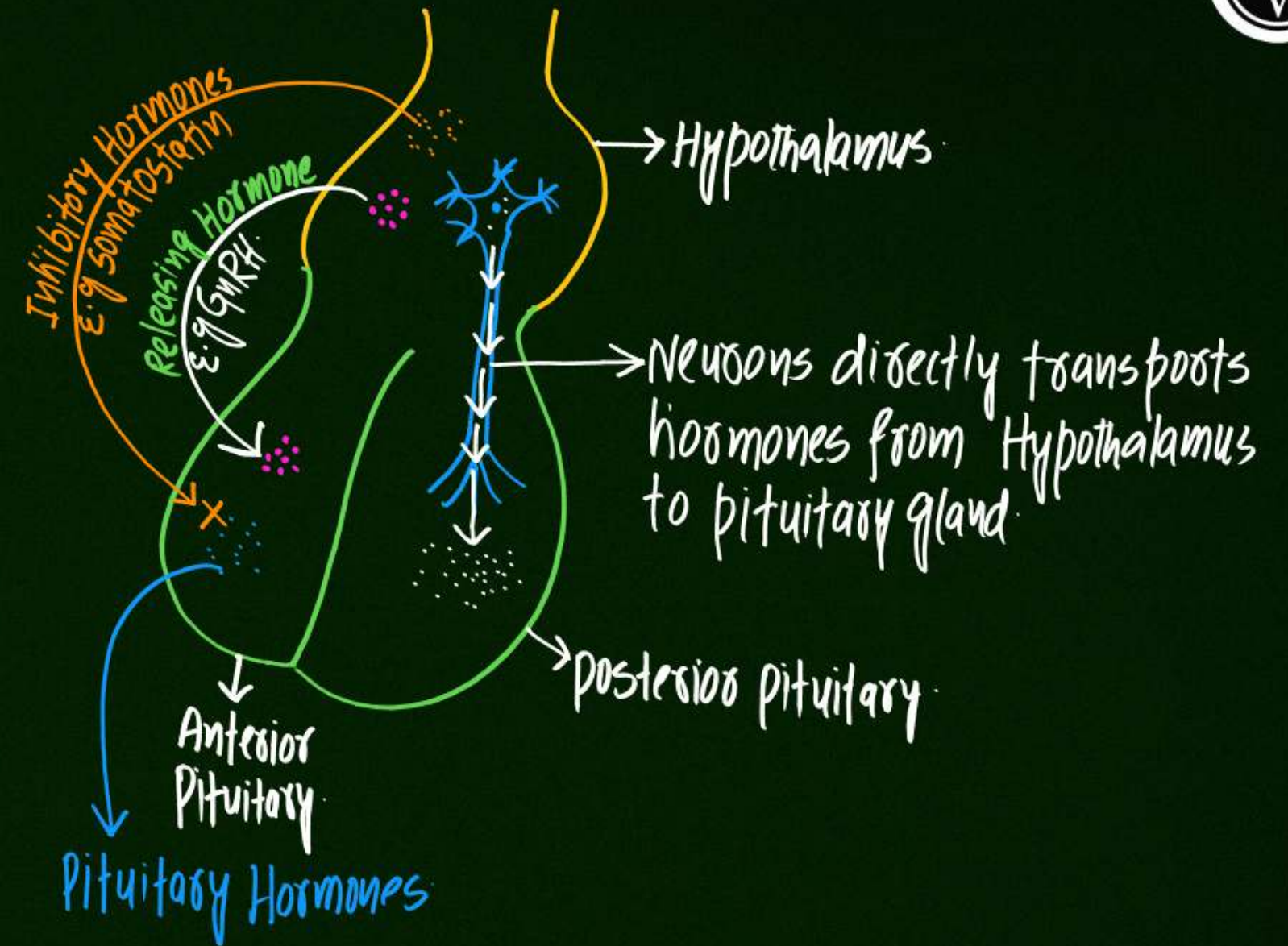
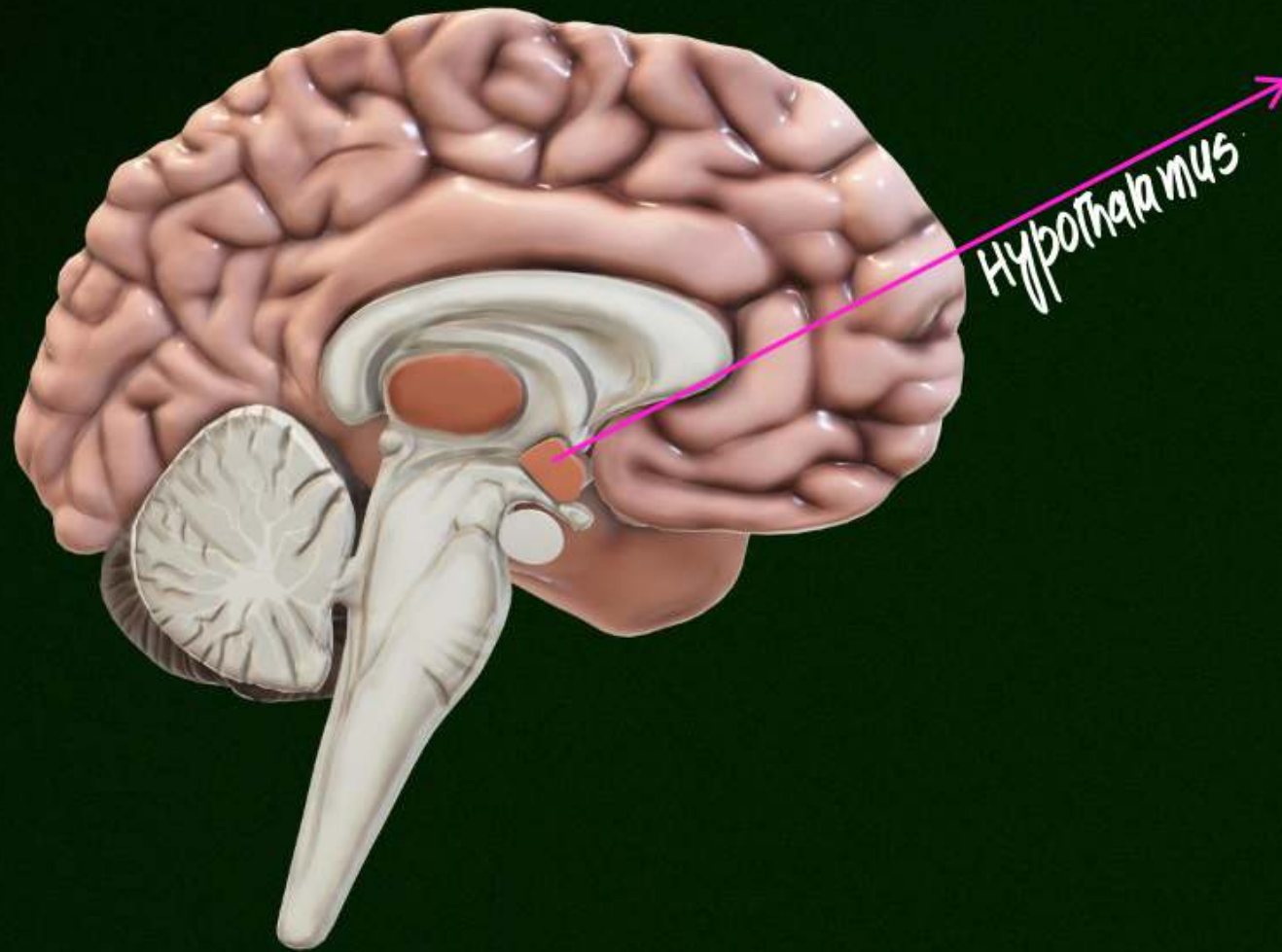


1. The Hypothalamus

Connection B/W Hypothalamus and Pituitary:



1. The Hypothalamus



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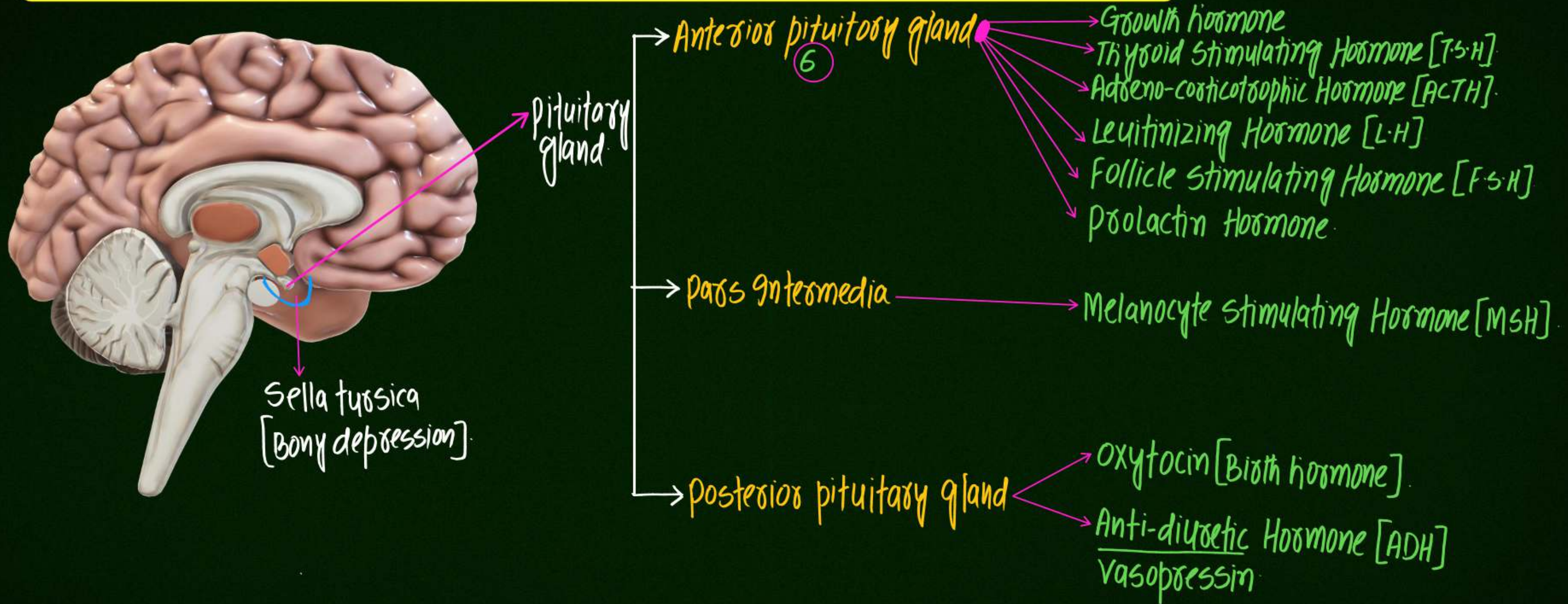
- Located in basal part of diencephalon (forebrain).
- Contains neurosecretory cells called nuclei that produce hormones.

Hormone Functions: They regulate synthesis and secretion of pituitary hormones.

Types of Hypothalamic Hormones:

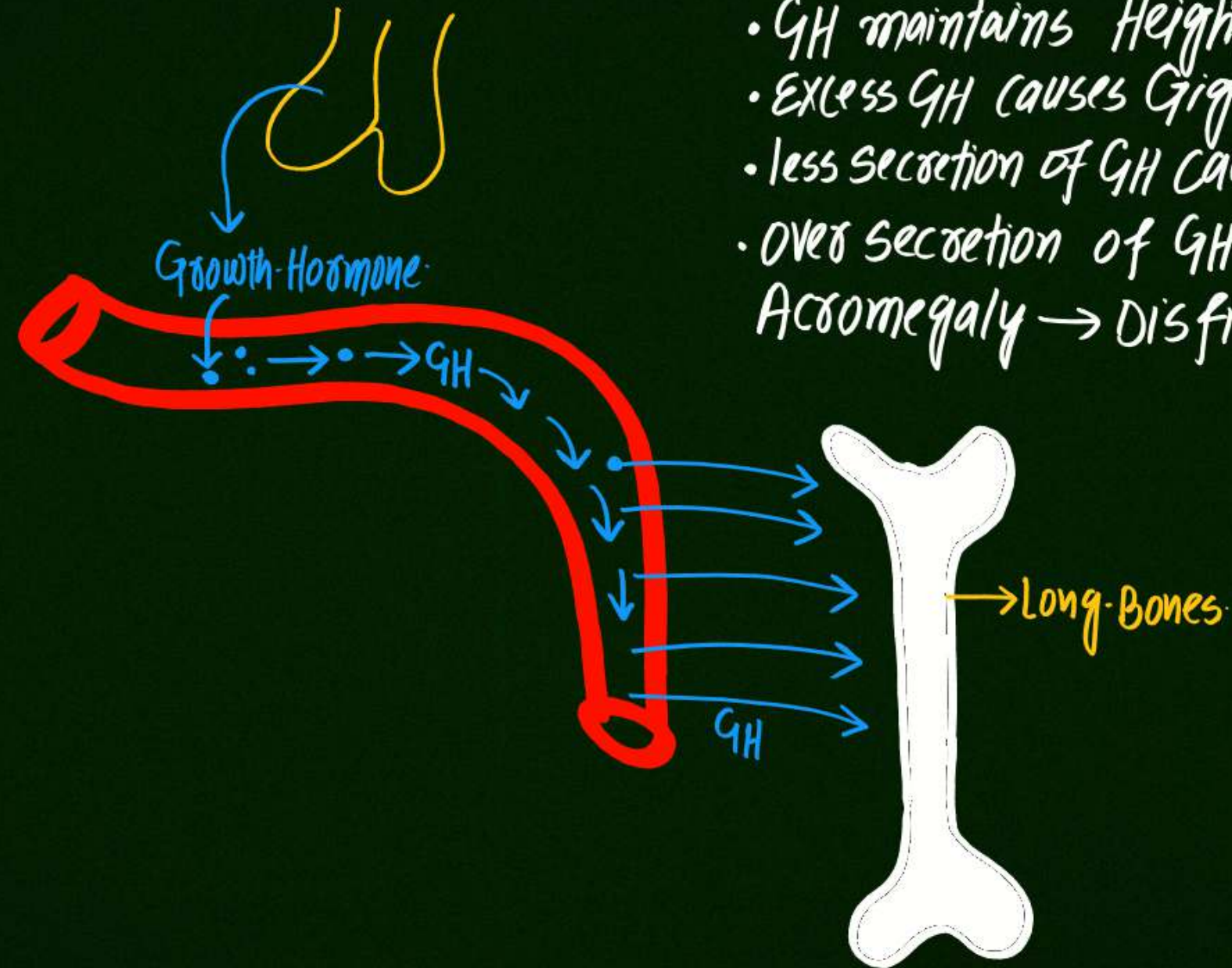
- Releasing Hormones (RH):** Stimulate secretion of pituitary hormones. Example: GnRH (Gonadotropin Releasing Hormone) stimulates release of gonadotrophins from pituitary.
 - Inhibiting Hormones (IH):** Inhibit secretion of pituitary hormones. Example: Somatostatin inhibits growth hormone release from pituitary.
- **Hormone Transport:**
 - Hormones from hypothalamus are released by neurons & travel via axons then enter portal circulatory system & reach to anterior pituitary.
 - Posterior pituitary gland is under direct neural regulation by the hypothalamus.

2. Pitutary Gland (Master Endocrine Gland)



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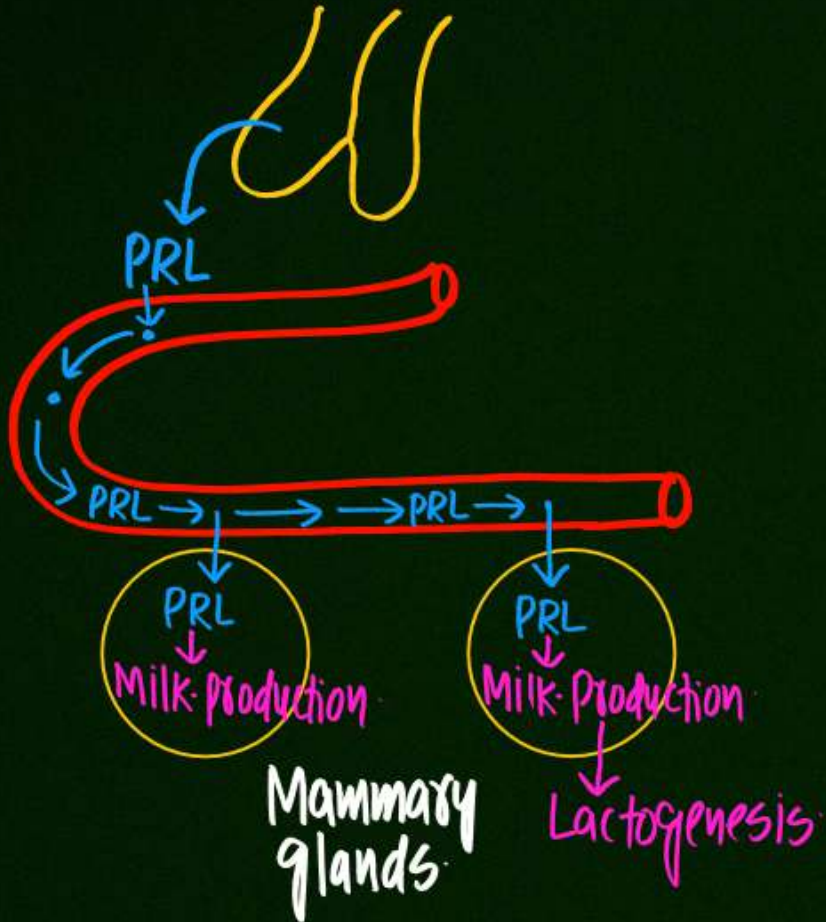
(i). Growth hormone (GH)



- GH maintains Height.
- Excess GH causes Gigantism (Tallness).
- less secretion of GH causes Dwarfism (short stunted).
- Over secretion of GH in adults causes Acromegaly → Disfigurement of body.

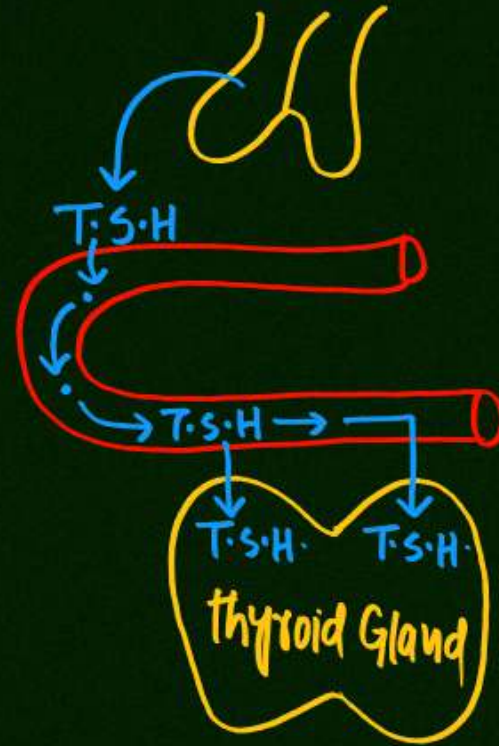
2. Pitutary Gland (Master Endocrine Gland)

(ii). Prolactin (PRL)



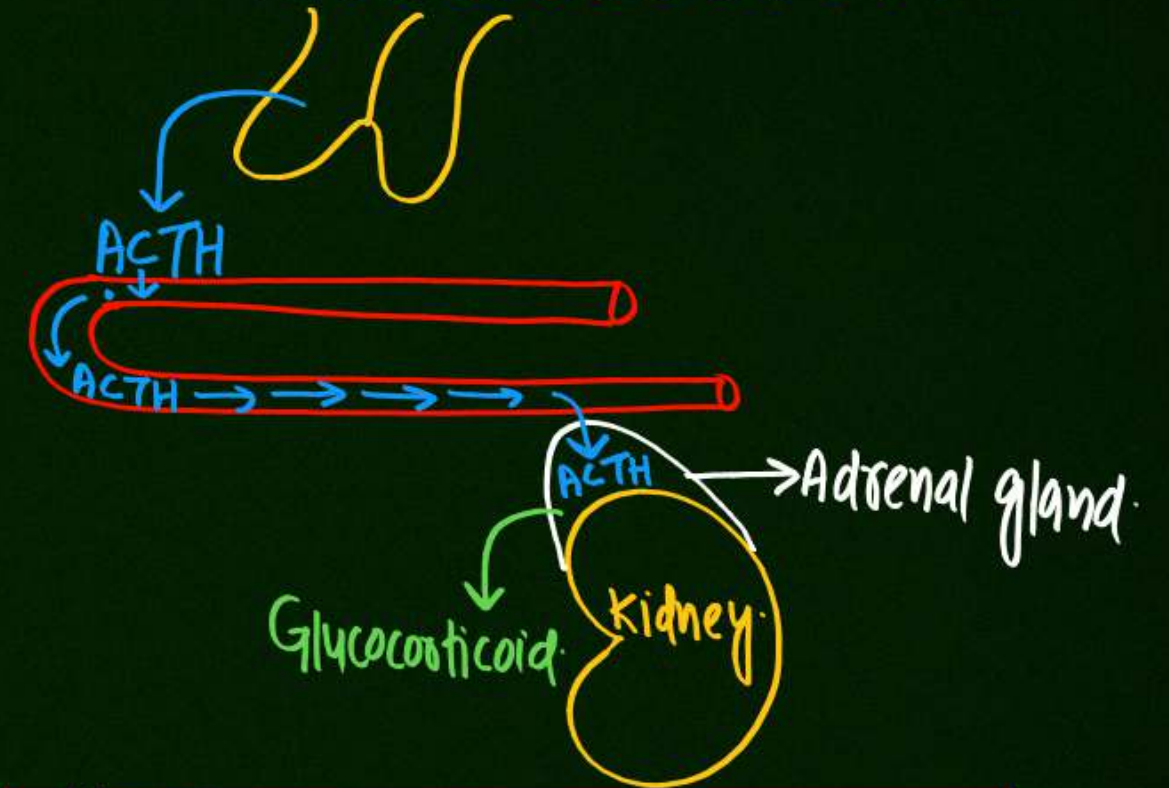
• Prolactin hormone stimulates Breasts to produce milk [lactogenesis].

(iii). TSH



• TSH stimulates thyroid gland to secrete its own Hormones

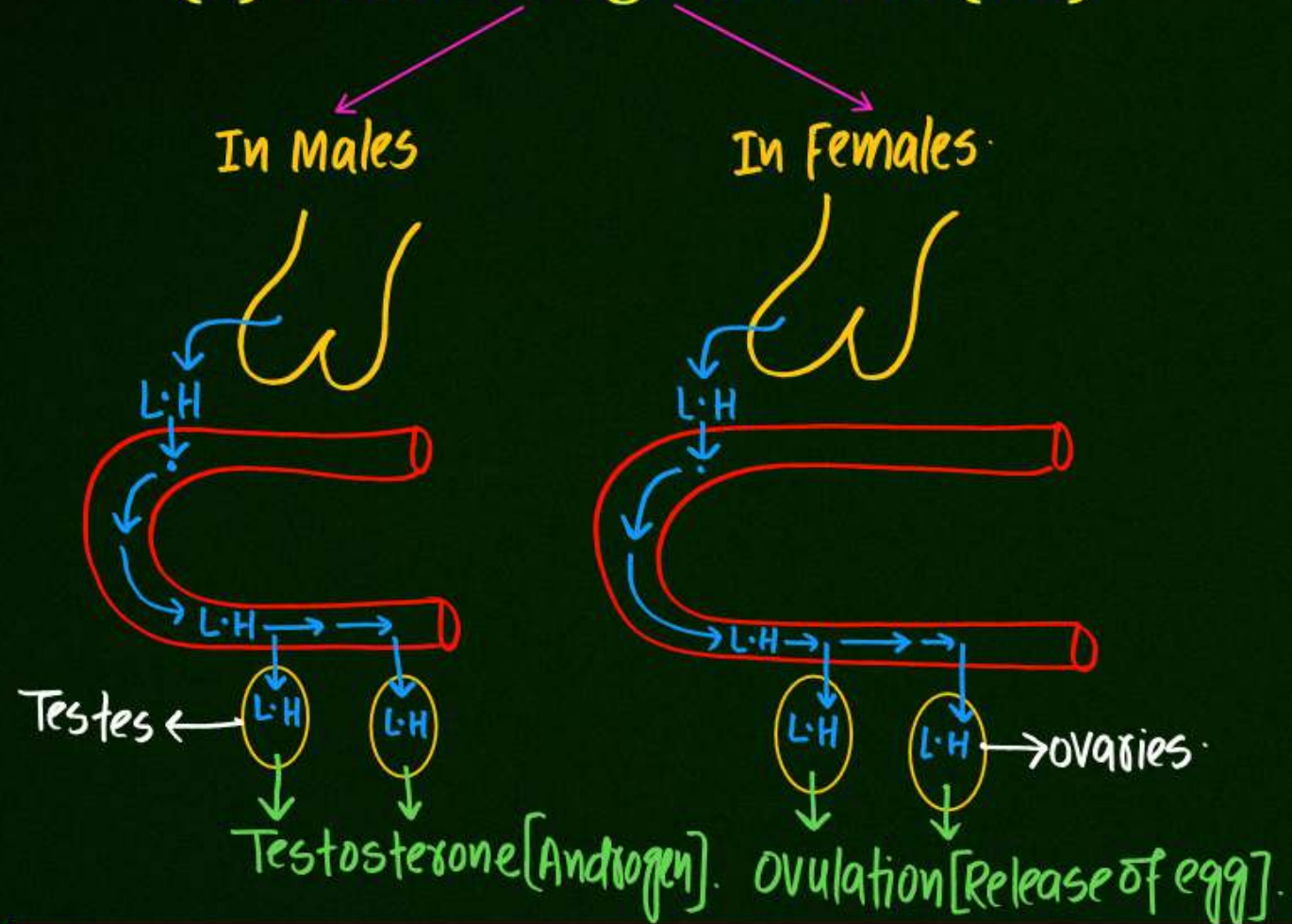
(iv). Adrenocorticotrophic hormone (ACTH)



• ACTH stimulates adrenal cortex to secrete glucocorticoids.

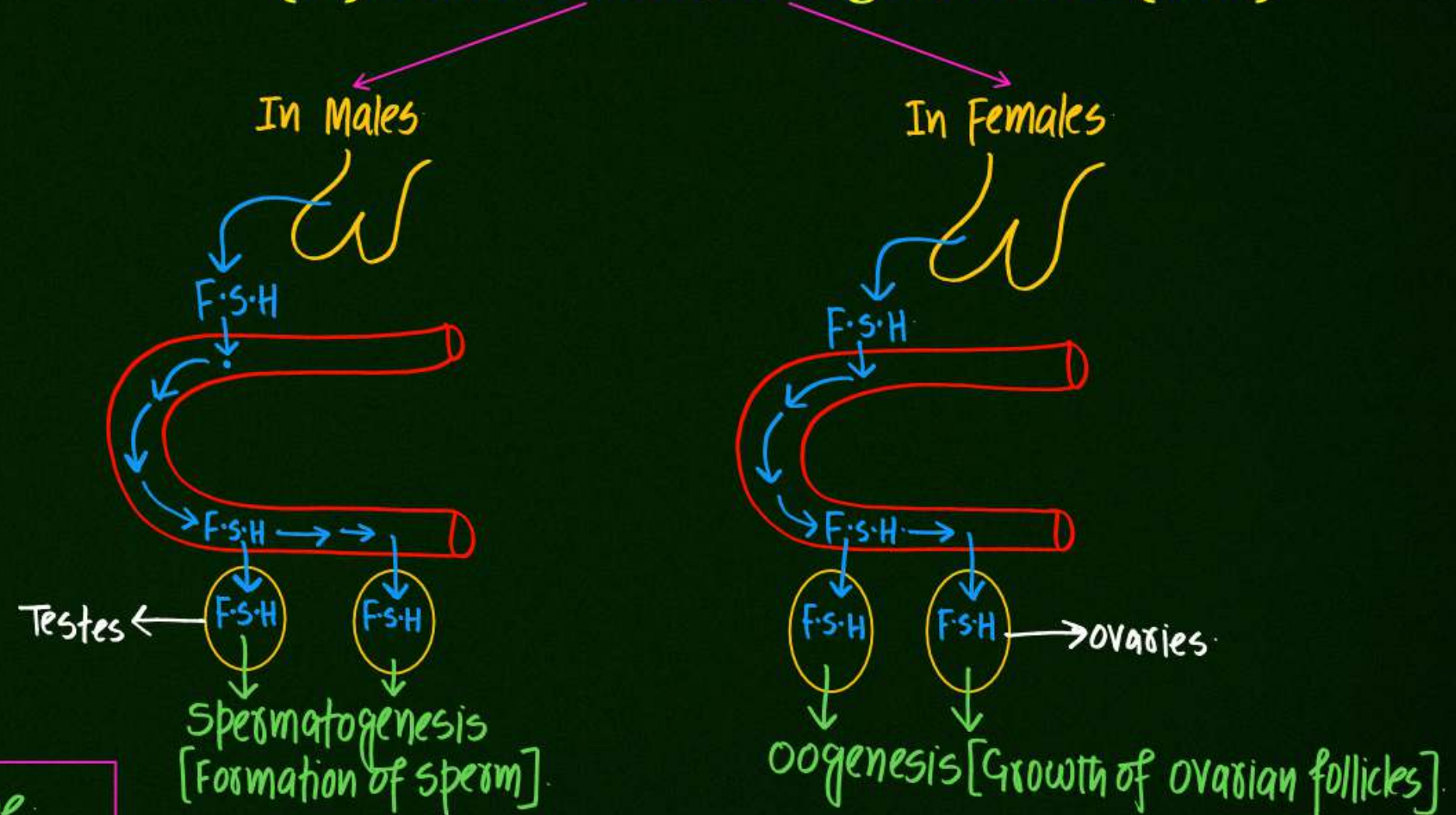
2. Pituitary Gland (Master Endocrine Gland)

(v). Luteinising hormone (LH)



- In Males L.H Stimulates Testes to Secrete Testosterone.
- In Females L.H Stimulates ovaries to release egg [ovulation]

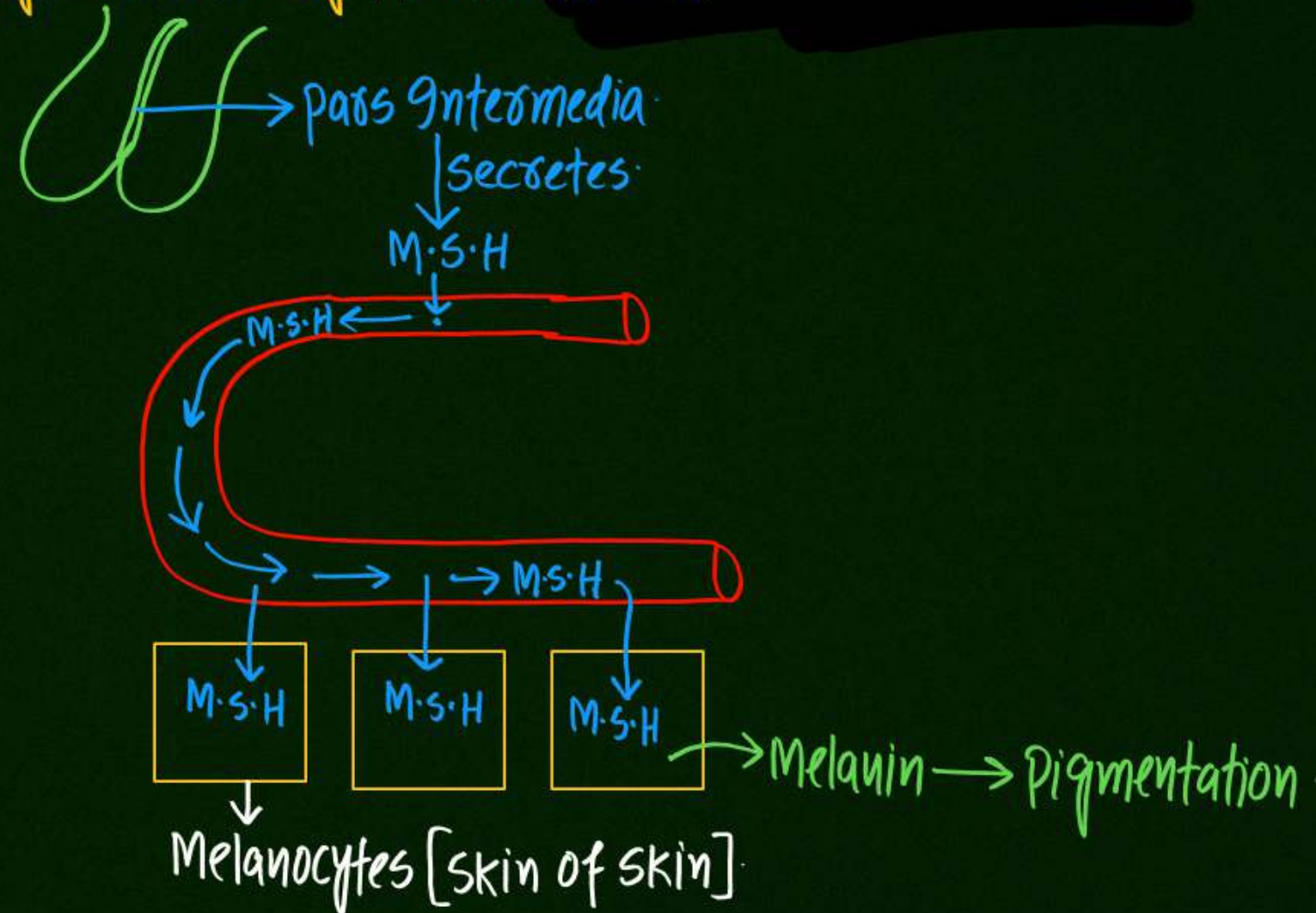
(vi). Follicle stimulating hormone (FSH)



- In males F.s.H Stimulates Testes for Spermatogenesis
- In Females F.s.H Stimulates ovaries for growth of ovarian follicles

2. Pitutary Gland (Master Endocrine Gland)

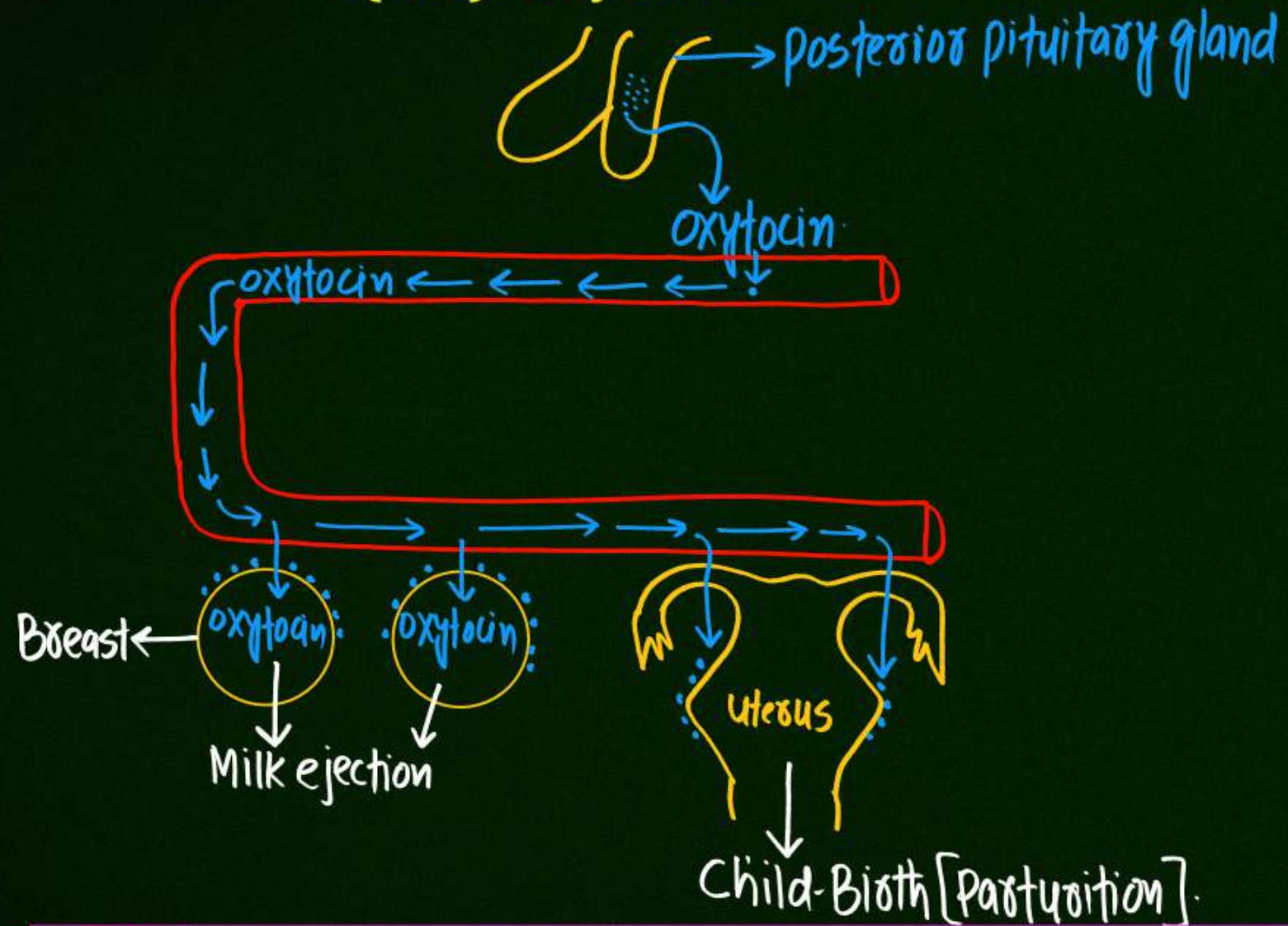
(vii) Melanocyte Stimulating Hormone [M.S.H.]



- MSH Stimulates Melanocytes of skin to secrete Melanin which causes Pigmentation.

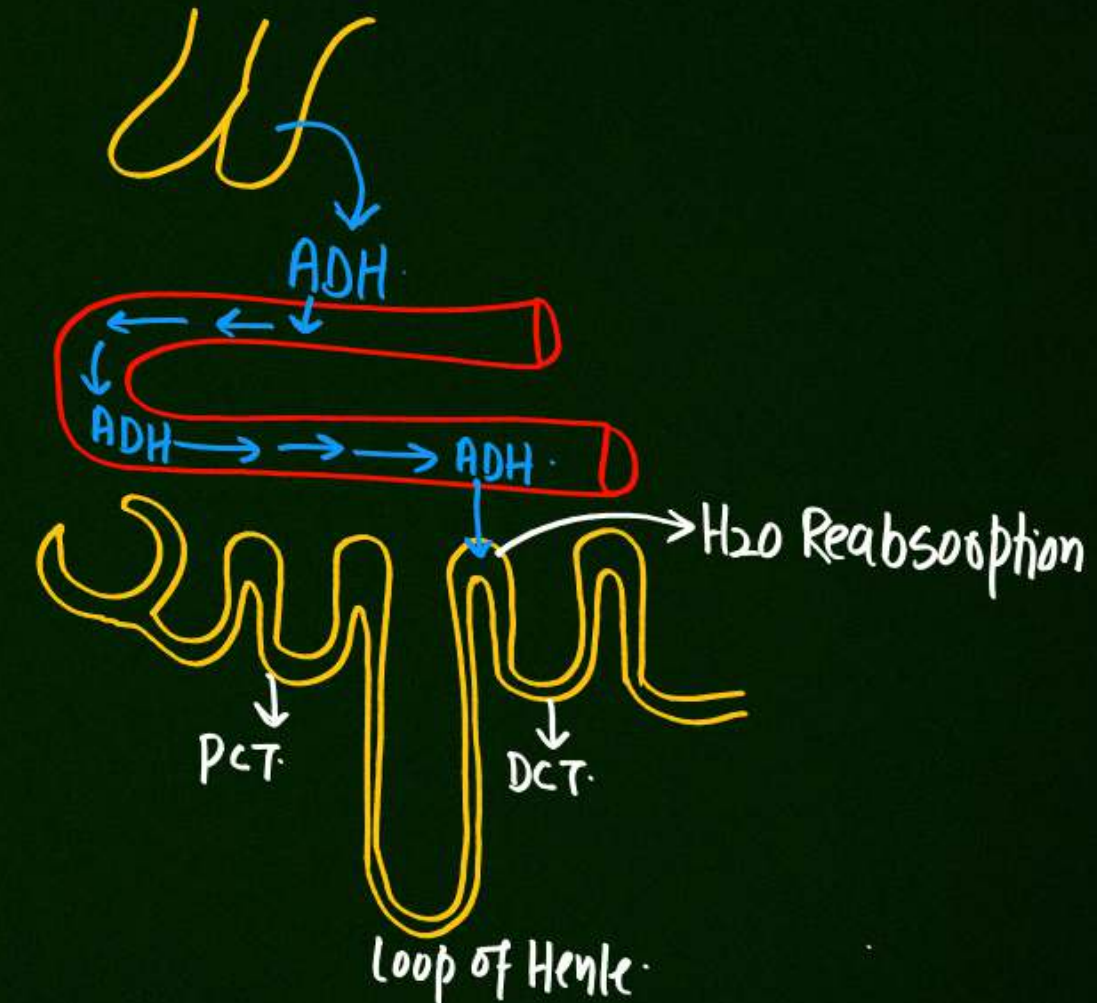
2. Pitutary Gland (Master Endocrine Gland)

(viii). Oxytocin



- Oxytocin acts on smooth muscles and stimulates contraction like breasts for milk ejection and contracts uterus for child birth [Birth hormone].

(ix). Vasopressin [ADH]



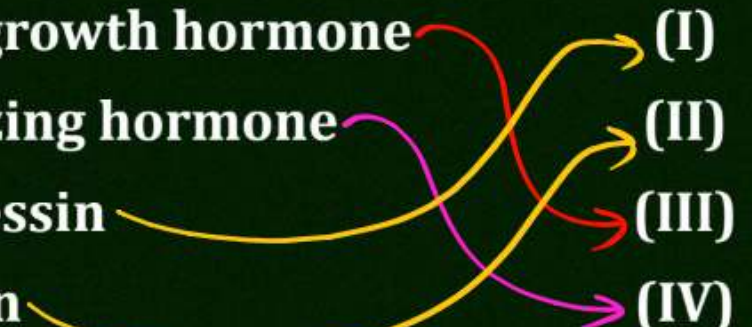
- ADH helps in H₂O reabsorption from DCT of Nephron and prevents from Diuresis.
- Less secretion of ADH causes Diuresis [Diluted Urine] and is known as **diabetic insipidus**.

Kaun Banayga Nabkaar [RENEET-2026]

Match List I with List II:

List I

List II

- | | |
|---------------------------|--|
| (A) Excess growth hormone | (I) Reabsorption of water and electrolytes in kidney |
| (B) Luteinizing hormone | (II) Contraction of uterus child birth |
| (C) Vasopressin | (III) Acromegaly |
| (D) Oxytocin | (IV) Ovulation |
- 

Choose the correct answer from the options given below:

A A-III, B-IV, C-II, D-I ✗

B A-III, B-IV, C-I, D-II ✓

C A-II, B-IV, C-I, D-III ✗

D A-IV, B-III, C-I, D-II ✗

Kaun Banayga Nabkaar [RENEET-2026]

Which of the following statements related to pituitary gland are correct?

- ✓ (a) It is divided anatomically into adenohypophysis and neurohypophysis ✓
- ✓ (b) It secretes follicle stimulating hormone
- ✓ (c) It secretes melanocyte stimulating hormone
- ✗ (d) It does not secrete prolactin

Choose the correct answer from the options given below:

A (a) And (b) only ✗

B (a), (b) and (c) only ✓

C (c) and (d) only ✗

D (b) and (c) only ✗

Kaun Banayga Nabkaar [NEET-2025]

Which of the following hormones released from the pituitary is actually synthesized in the hypothalamus?

A

Anti-diuretic hormone (ADH) *Vasopressin*

B

Follicle-stimulating hormone (FSH) ✗

C

Adenocorticotrophic hormone (ACTH) ✗

D

Luteinizing hormone (LH) ✗

Thank
You